Program: **Diploma in Integrated Circuit Design and Fabrication**	
Course code: **3372**	Course Title: **Solid State Devices**
Semester: **3**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **3 (L: 3 T: 0 P :0)**	Periods per semester: **45**

Course Objective:

- To provide an insight into the physics of solid State Device

Course Prerequisites:

Topic	Course code	Course name	semester
Basics of Electronic Devices	2041	Basic Electronics	2
Basics of Semiconductor physics	2003	Applied Physics II	2
Basic engineering mathematics principles & Theorems	1002 2002	Mathematics 1 Mathematics 2	1 2

Course outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the types of semiconductors, Fermi -Dirac distribution and carrier concentration	11	**Understanding**
CO2	Explain carrier transport, excess carrier generation and recombination mechanisms in semiconductors.	11	**Understanding**
CO3	Describe energy band diagram of P-N junction and break down mechanism in P-N junction	10	**Understanding**
CO4	Explain Metal-Semiconductor contacts, BJT,FET and its characteristics	11	**Understanding**
	Series Test	2	

CO-PO mapping

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3	3					
CO3	3	3					
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the type of semiconductors, fermi-Dirac distribution and carrier concentration.**		
M1.01	Illustrate the energy bands in intrinsic and extrinsic semiconductors.	2	Understanding
M1.02	Illustrate energy -momentum relation for electrons in solids.	2	Understanding
M1.03	Explain fermi -Dirac distribution function and equilibrium concentration of electron and holes.	5	Understanding
M1.04	Explain temperature dependence of carrier concentration	2	Understanding
Contents: Elemental and compound semiconductor materials: Energy bands in solids, intrinsic and extrinsic semiconductors-valence band model, n-type, p type, Variation of Energy band gap with alloy composition. Energy -momentum relation for electrons in solids, effective mass. Fermi- Dirac distribution. Equilibrium and steady state conditions, Equilibrium concentration of electrons and holes (no derivation.) Temperature dependence of carrier concentration.			
CO2	**Explain carrier transport, excess carrier generation and recombination mechanisms in semiconductors**		
M2.01	Explain carrier transport mechanism	3	Understanding
M2.02	Explain high field effect and Hall effect	3	Understanding
M2.03	Explain excess carrier generation in semiconductors.	3	Understanding
M2.04	Explain the recombination mechanisms in semiconductor	2	Understanding
	Series test 1	1	

Contents: Carrier transport in semiconductors-drift,diffusion,conductivity and mobility, variation of mobility with temperature and doping, High Field effects, Hall effect. Excess carriers in semiconductors-photogeneration, low level and high level injections, Recombination mechanisms-Indirect recombination via deep energy levels in the band gap, Auger recombination, Trap- aided Auger recombination.			
CO3	**Describe energy band diagram of P-N junction and break down mechanism in P-N junction.**		
M3.01	Illustrate the equilibrium energy band diagrams of P-N Junction	3	Understanding
M3.02	Describe current flow through a diode	2	Understanding
M3.03	Explain V-I characteristics of diode and its equation	2	Understanding
M3.04	Explain the breakdown mechanism in P-N junctions	3	Understanding
Contents: P-N junction under thermal equilibrium, equilibrium energy band diagram, biasing of P-N junction, qualitative description of current flow through a diode, ideal diode equation (no derivation), V-I characteristics of real diode, Breakdown mechanism in P-N junctions- Zener, Avalanche breakdown, Punched through diode, application of breakdown diodes.			
CO4	**Explain metal - semiconductor contacts, BJT, FET and its characteristics**		
M4.01	Explain metal -semiconductor contacts	3	Understanding
M4.02	Explain current components of BJT and Static I-V characteristics of CB and CE configurations	3	Understanding
M4.03	Describe BJT breakdown mechanisms	2	Understanding
M4.04	Explain the principle and operation of JFET	3	Understanding
	Series test 2	1	
Contents: Metal Semiconductor contacts, Energy band diagram of Ohmic and Rectifying Contacts, Current Voltage characteristics, Comparison with PN Junction Diode. Bipolar junction transistor-Current components, Base width modulation, Avalanche multiplication in collector- base junction, Punch Through, Base resistance, Static I-V characteristics of CB and CE configurations. Field Effect Transistors: JFET - Principle of operation.			

Text/reference:

T/R	Book Title/Author
T1	Suresh Babu V, Solid State Device Technology, Sanguine, 2005.
T2	Somanathan Nair, Solid State Devices, Prentice Hall of India, 2/e, 2010
R1	Streetman and Banerjee, Solid State electronic Devices, Prentice Hall of India,6/e 2010
R2	Tyagi M.S., Introduction to Semiconductor Materials and Devices, Wiley India, S/e, 2008.
R3	Wamer and Grung, Semiconductor Device Electronics, Holt Rinehart and Winston, 1991.
R4	Sze S.M., Physics of Semiconductor Devices, John Wiley, 3/e, 2005.

Online resources:

SL. No.	Website link
1	https://en.wikipedia.org/wiki/Fermi Dirac Statistics
2	https://www.electronics-tutorials.ws/diode/diode_3.html
3	https://en.wikipedia.org/wiki/Metal semiconductor junction